

# Scanning for European Bonds with the IBKR API

A practical guide to finding European government and corporate bonds through the IBKR market scanner. Every code

below comes from IBKR's scanner parameters XML, returned by `ib.reqScannerParametersAsync()`. If IBKR changes its scanner, fetch a fresh copy and check the codes against it.

The functions used here (`interactive_brokers_scan_bonds`, `quote_bonds`, `print_bonds_table`, ...) are documented in [Scanners](#).

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## 1. How a scan is built

An IBKR scan has four parts:

| Part                      | What it means                                      | Example                                            |
|---------------------------|----------------------------------------------------|----------------------------------------------------|
| <code>instrument</code>   | Which kind of security to scan                     | <code>BOND</code> , <code>BOND.GOV.T.NON-US</code> |
| <code>locationCode</code> | Which market or venue to scan                      | <code>BOND.GOV.T.EU.EURONEXT</code>                |
| <code>scanCode</code>     | How results are <b>sorted</b> (and loosely chosen) | <code>HIGH_BOND_ASK_YIELD_ALL</code>               |
| <code>filter tags</code>  | Conditions every result must meet                  | <code>currencyLike=EUR</code>                      |

The scan code only decides the order. The filters do the real narrowing. A scan returns at most 50 rows, so tight filters matter: otherwise you only see the top 50 of an arbitrary universe.

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## 2. Where European bonds live

*Instruments*

| instrument        | Covers                      |
|-------------------|-----------------------------|
| BOND              | Corporate bonds (worldwide) |
| BOND.GOV.T.NON-US | Non-US sovereign bonds      |

(BOND.GOV.T, BOND.AGENCY, BOND.MUNI and BOND.CD are US-only.)

## Locations

| locationCode           | Name                       | Use with          |
|------------------------|----------------------------|-------------------|
| BOND.WW                | Worldwide corporate bonds  | BOND              |
| BOND.US                | US corporate bonds         | BOND              |
| BOND.EU.EURONEXT       | Euronext corporate bonds   | BOND              |
| BOND.EU.EBS            | SIX Swiss corporate bonds  | BOND              |
| BOND.GOV.T.NON-US      | All non-US sovereign bonds | BOND.GOV.T.NON-US |
| BOND.GOV.T.US.NON-US   | Non-US govt bonds          | BOND.GOV.T.NON-US |
| BOND.GOV.T.EU.EURONEXT | Euronext govt bonds        | BOND.GOV.T.NON-US |
| BOND.GOV.T.EU.EBS      | SIX Swiss govt bonds       | BOND.GOV.T.NON-US |
| BOND.GOV.T.HK.SEHK     | Hong Kong bonds            | BOND.GOV.T.NON-US |

BOND.WW contains BOND.US, BOND.EU.EURONEXT and BOND.EU.EBS. BOND.GOV.T.NON-US contains the four govt locations below it.

### Which one to pick:

- **Broadest search (recommended starting point):** use a parent location (BOND.WW or BOND.GOV.T.NON-US), then narrow it with currencyLike and issuerCountryIs. Many European bonds are available outside Euronext and SIX, so a venue-specific location can miss them.
- **Only bonds on a particular exchange:** use BOND.EU.EURONEXT, BOND.EU.EBS, BOND.GOV.T.EU.EURONEXT or BOND.GOV.T.EU.EBS.

## *Two restrictions on the exchange locations*

The XML marks the Euronext and SIX locations with:

- **access="restricted;s=..."**: they need the matching market-data subscription on your account. Without it, the scan returns nothing or an error.
  - **rawPriceOnly="yes"** (the two *corporate* exchange locations): only prices are available, so yield, spread and duration filters may not work there. Filter on price (`bondAsk...`, `bondBidOrAsk...`) instead.
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### 3. Scan codes (sort order)

| scanCode                           | Sorts by                     | Corp | Govt |
|------------------------------------|------------------------------|------|------|
| HIGH_BOND_ASK_YIELD_ALL            | Highest ask yield            | ✓    | ✓    |
| LOW_BOND_BID_YIELD_ALL             | Lowest bid yield             | ✓    | ✓    |
| HIGH_BOND_SPREAD_ALL               | Widest spread                | ✓    | ✓    |
| LOW_BOND_SPREAD_ALL                | Tightest spread              | ✓    | ✓    |
| HIGH_COUPON_RATE                   | Highest coupon               | ✓    | ✓    |
| LOW_COUPON_RATE                    | Lowest coupon                | ✓    | ✓    |
| NEAR_MATURITY_DATE                 | Soonest maturity             | ✓    | ✓    |
| FAR_MATURITY_DATE                  | Latest maturity              | ✓    | ✓    |
| BOND_CUSIP_AZ / _ZA                | Identifier, alphabetical     | ✓    | ✓    |
| HIGH_MOODY_RATING_ALL              | Best Moody's rating first    | ✓    |      |
| LOW_MOODY_RATING_ALL               | Worst Moody's rating first   | ✓    |      |
| HIGH_SP_RATING_ALL                 | Best S&P rating first        | ✓    |      |
| LOW_SP_RATING_ALL                  | Worst S&P rating first       | ✓    |      |
| MOST_ACTIVE                        | Volume                       | ✓    |      |
| TOP_TRADE_COUNT                    | Number of trades             | ✓    |      |
| TOP_PERC_GAIN / _LOSE              | Price change today           | ✓    |      |
| HIGH_/LOW_BOND_DEBT_2_EQUITY_RATIO | Issuer leverage              | ✓    |      |
| SCAN_esgScore_DESC / _ASC          | Issuer ESG score (Refinitiv) | ✓    |      |

The XML also has the other Refinitiv balance-sheet ratio scans and the individual ESG pillar scans, for corporates only.

## 4. Filters

Most numeric filters come in pairs, ...Above and ...Below. Pass them as strings.

## *Available for both corporates and sovereigns*

| Tag                             | Meaning / units                                                  |
|---------------------------------|------------------------------------------------------------------|
| currencyLike                    | EUR, GBP, CHF, USD, CAD, AUD, BRL, HKD                           |
| issuerCountryIs                 | ISO 2-letter code (see section 5)                                |
| maturityDateAbove / Below       | yyyymmdd only (IBKR rejects mm/yyyy and years)                   |
| couponRateAbove / Below         | Coupon in %                                                      |
| bondAskYieldAbove / Below       | Yield at the ask, in %                                           |
| bondBidYieldAbove / Below       | Yield at the bid, in %                                           |
| bondBidOrAskYieldAbove / Below  | Yield at bid or ask, in %                                        |
| bondAskAbove / Below            | Ask price (% of face value)                                      |
| bondBidOrAskAbove / Below       | Bid or ask price                                                 |
| bondAskSizeValueAbove / Below   | Size on the ask, <b>in thousands</b> of face value               |
| bondSpreadAbove / Below         | Spread (unit not stated in the XML; likely basis points)         |
| bondDurationAbove / Below       | Duration                                                         |
| bondConvexityAbove / Below      | Convexity                                                        |
| bondAmtOutstandingAbove / Below | Issue size outstanding, <b>in millions</b> of face value         |
| bondInitialSizeAbove / Below    | Minimum order size (face value)                                  |
| bondIncrementSizeAbove / Below  | Order size increment (face value)                                |
| bondVarCouponRateIs             | true = only floaters, false = exclude floaters                   |
| excludeConvertible              | true to exclude convertibles                                     |
| bondValidBidOrAskOnly           | true <b>includes</b> bonds with no quote (see the warning below) |

**Warning: bondValidBidOrAskOnly does the opposite of what its name suggests.** The XML labels it "Include bonds without quotes" and says unquoted bonds are excluded by default. Leave it out to get only bonds you can trade now. Set it to true only to see the full universe, including bonds with no quote.

## *Corporates only (BOND)*

| Tag                                                                         | Meaning / values                                                           |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| bondCreditRating                                                            | highGrade (investment grade) or highYield                                  |
| moodyRatingAbove / Below                                                    | AAA AA1 AA2 AA3 A1 A2 A3 BAA1 BAA2 BAA3 BA1 ... B3 CAA1 ... C NR           |
| spRatingAbove / Below                                                       | AAA AA+ AA AA- A+ A A- BBB+ BBB BBB- BB+ ... D NR                          |
| ratingsRelation                                                             | How Moody's and S&P filters combine: or, and, xand                         |
| bondIssuerLike                                                              | Issuer name, partial match (e.g. Siemens)                                  |
| bondStkSymbolIs                                                             | Ticker of the issuer's stock (e.g. SAP)                                    |
| bondCallableIs                                                              | true only / false exclude                                                  |
| bondNextCallDateAbove / Below                                               | Call protection: a date, presumably yyyyymmdd like maturity (untested)     |
| bondPaymentFreqIs                                                           | 1 annual, 2 semi-annual, 4 quarterly, 12 monthly, 0 at maturity            |
| bondDefaultedIs                                                             | false to exclude defaulted bonds                                           |
| bondTradingFlatIs                                                           | false to exclude bonds trading without accrued interest (often distressed) |
| bondExchListedIs                                                            | true only exchange-listed                                                  |
| bondStkMarketCapAbove / Below                                               | Issuer equity market cap, in millions                                      |
| bondDebtOutstandingAbove / Below                                            | Issuer total debt, in millions                                             |
| bondDebt2BookRatioAbove / Below                                             | Issuer debt/book (Refinitiv)                                               |
| esgScoreAbove / Below                                                       | Issuer ESG score (Refinitiv); pillar scores also available                 |
| priceAbove/Below, volumeAbove, changePercAbove/Below, tradeCountAbove/Below | Trading activity                                                           |

**Sovereigns have no rating filters or rating scans.** To restrict by credit quality, filter by issuerCountryIs instead (e.g. scan only DE, NL, AT, FI for AAA/AA euro sovereigns).

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## 5. European country codes

The issuerCountryIs values for Europe, taken from the XML:

| Code | Country        | Code | Country    | Code | Country     |
|------|----------------|------|------------|------|-------------|
| AT   | Austria        | FR   | France     | NL   | Netherlands |
| BE   | Belgium        | DE   | Germany    | NO   | Norway      |
| BG   | Bulgaria       | GR   | Greece     | PL   | Poland      |
| HR   | Croatia        | HU   | Hungary    | PT   | Portugal    |
| CY   | Cyprus         | IE   | Ireland    | RO   | Romania     |
| CZ   | Czech Rep.     | IT   | Italy      | SK   | Slovakia    |
| DK   | Denmark        | LV   | Latvia     | SI   | Slovenia    |
| EE   | Estonia        | LT   | Lithuania  | ES   | Spain       |
| FI   | Finland        | LU   | Luxembourg | SE   | Sweden      |
| GB   | United Kingdom | MT   | Malta      | CH   | Switzerland |

issuerCountryIs takes one country per scan. To cover several countries, run one scan per country (section 7).

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## 6. Running a scan

interactive\_brokers\_scan\_bonds lives in  
clients/interactive\_brokers/scanners/fixed\_income.py:

```

from clients.interactive_brokers.client import connect, disconnect
from clients.interactive_brokers.scanners.fixed_income import
interactive_brokers_scan_bonds

ib = await connect()
try:
    bonds = await interactive_brokers_scan_bonds(ib,
instrument="BOND.GOV.T.NON-US", location="BOND.GOV.T.NON-US",

scan_code="HIGH_BOND_ASK_YIELD_ALL", issuerCountryIs="DE",
currencyLike="EUR")
finally:
    disconnect(ib)

```

Pass any filter from section 4 as a keyword argument; values are converted to strings for you. Tags that are not in

BondFilters still work but are flagged by type checkers.

interactive\_brokers\_scan\_bonds returns a list of Bond objects with ISIN, coupon, maturity and the other details. Section 8 shows how to add prices and yields; [Scanners](#) explains where each field comes from.

## 7. Recipes

### *7.1 German Bunds, 2 to 10 years, highest yield first*

```

await interactive_brokers_scan_bonds(
    ib, instrument="BOND.GOV.T.NON-US", location="BOND.GOV.T.NON-US",
scan_code="HIGH_BOND_ASK_YIELD_ALL",
    issuerCountryIs="DE",
    currencyLike="EUR",
    maturityDateAbove="20281002",    # 2 years from 2 Oct 2026
    maturityDateBelow="20361002",    # 10 years
    bondVarCouponRateIs="false",
)

```

### *7.2 Euro-area periphery sovereigns paying over 3%*



```

results = {}
for country in ["IT", "ES", "PT", "GR"]:
    results[country] = await interactive_brokers_scan_bonds(
        ib, instrument="BOND.GOV.NON-US", location="BOND.GOV.NON-US",
        scan_code="HIGH_BOND_ASK_YIELD_ALL",
        issuerCountryIs=country,
        currencyLike="EUR",
        bondAskYieldAbove=3,
        maturityDateBelow="20331002",    # 7 years
    )

```

### *7.3 Short-dated core-euro govts (cash-like ladder)*

```

await interactive_brokers_scan_bonds(
    ib, instrument="BOND.GOV.NON-US", location="BOND.GOV.NON-US",
    scan_code="NEAR_MATURITY_DATE",
    issuerCountryIs="NL",
    currencyLike="EUR",
    maturityDateBelow="20271231",
    bondAmtOutstandingAbove=5_000,    # >= EUR 5bn outstanding, i.e. liquid
    lines
)

```

### *7.4 Investment-grade EUR corporates, plain vanilla, 3 to 7 years*

```

await interactive_brokers_scan_bonds(
    ib, instrument="BOND", location="BOND.WW",
    scan_code="HIGH_BOND_ASK_YIELD_ALL",
    currencyLike="EUR",
    bondCreditRating="highGrade",
    maturityDateAbove="20291002",    # 3 years
    maturityDateBelow="20331002",    # 7 years
    bondCallableIs="false",
    excludeConvertible="true",
    bondVarCouponRateIs="false",
    bondDefaultedIs="false",
    bondAmtOutstandingAbove=500,    # >= EUR 500m issue size
)

```

### *7.5 A-rated or better from both agencies, French issuers*

```

await interactive_brokers_scan_bonds(
    ib, instrument="BOND", location="BOND.WW",
    scan_code="HIGH_SP_RATING_ALL",
    issuerCountryIs="FR",
    currencyLike="EUR",
    spRatingAbove="A-",
    moodyRatingAbove="A3",
    ratingsRelation="and",
)

```

Check the direction with a bond you know before relying on it. "Above" should mean "this rating or better", but the XML does not say.

## 7.6 EUR high yield with guard rails

```

await interactive_brokers_scan_bonds(
    ib, instrument="BOND", location="BOND.WW",
    scan_code="HIGH_BOND_ASK_YIELD_ALL",
    currencyLike="EUR",
    bondCreditRating="highYield",
    spRatingAbove="BB-",           # stay in the upper part of high yield
    bondAskYieldAbove=5,
    bondAskYieldBelow=10,         # very high yields usually mean
    distress
    bondDurationBelow=4,
    bondDefaultedIs="false",
    bondTradingFlatIs="false",
    bondCallableIs="false",
)

```

## 7.7 Bonds you can buy in small amounts

Many European corporate bonds have a EUR 100,000 minimum denomination. To leave those out:

```

await interactive_brokers_scan_bonds(
    ib, instrument="BOND", location="BOND.WW",
    scan_code="HIGH_BOND_ASK_YIELD_ALL",
    currencyLike="EUR",
    bondCreditRating="highGrade",
    bondInitialSizeBelow=100,     # minimum order below 100 units
)

```

The XML does not state the unit of `bondInitialSize`. It appears to match IBKR's order quantity, the same unit as

`Bond.minimum_size`, and that unit is not currency: a what-if order on a Bund showed one unit = €1,000 of face value.

So `minimum_size = 1` means about €1,000 and `100` about €100,000. Check a bond's unit in TWS before trading.

## *7.8 One issuer's whole curve*

```
await interactive_brokers_scan_bonds(
    ib, instrument="BOND", location="BOND.WW",
    scan_code="FAR_MATURITY_DATE",
    bondIssuerLike="Volkswagen",
    currencyLike="EUR",
)
```

## *7.9 Swiss franc bonds on SIX*

```
# Sovereigns: yield filters are available here
await interactive_brokers_scan_bonds(ib, instrument="BOND.GOV.T.NON-US",
    location="BOND.GOV.T.EU.EBS", scan_code="HIGH_BOND_ASK_YIELD_ALL",
    currencyLike="CHF")

# Corporates: SIX corporate data is price-only, so filter and sort on price
await interactive_brokers_scan_bonds(
    ib, instrument="BOND", location="BOND.EU.EBS",
    scan_code="HIGH_COUPON_RATE",
    currencyLike="CHF",
    bondAskBelow=100,          # trading below par
    maturityDateBelow="20311002", # 5 years
)
```

## *7.10 Low-carbon EUR corporates*

```
await interactive_brokers_scan_bonds(
    ib, instrument="BOND", location="BOND.WW",
    scan_code="SCAN_esgEmissionsScore_DESC",
    currencyLike="EUR",
    bondCreditRating="highGrade",
    esgEmissionsScoreAbove=70,
)
```

---

## 8. Prices and yields

`quote_bonds` returns the same bonds with a quote (clean price, time, live flag);  
`print_bonds_table` and `print_bond`  
show them with their yields:

```
from clients.interactive_brokers.scanners.fixed_income import quote_bonds
from clients.interactive_brokers.scanners.printing import print_bond,
print_bonds_table

quoted = await quote_bonds(ib, bonds)
print_bonds_table(quoted, title="GERMAN GOVERNMENT BONDS")
print_bond(quoted[0])

bond = quoted[0]
bond.dirty_price() # 98.193 for OBL 2 1/2 04/16/31 at 97.035 clean
bond.yield_to_maturity() # 3.21 (%)
bond.yield_without_reinvestment() # 3.04 (%), coupons kept as cash
```

Prices are requested as delayed-frozen data: real-time where you subscribe, otherwise delayed or the last close  
(`quote.live` is `False`). Without any price, quote and every yield are `None`.

Coupons are treated as annual, because IBKR does not report the payment frequency. That is right for German and most  
other euro government bonds; for semi-annual payers (Italian BTPs, UK gilts, most USD bonds) the computed yield is  
slightly off. The full calculation rules are in [Scanners](#).

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## 9. Pitfalls

- **Empty results.** Remove filters one at a time. The usual causes are a missing market-data subscription for the  
location, a yield filter on a price-only location, or filters that don't match each other.

- **Unsupported filters are dropped without warning.** A filter that doesn't apply to the instrument, such as a rating filter on sovereigns, is usually ignored rather than rejected, so the results can look filtered when they aren't. Each instrument's supported filters are in its `<filters>` list in the scanner parameters XML.
  - **50-row cap.** If a scan returns exactly 50 rows, you are probably missing bonds. Tighten the filters or split the scan by country or maturity range.
  - **Scanner yields are snapshots.** For the bonds you shortlist, check yields against live market data.
  - **Units differ by filter.** Issue size is in millions and quoted size is in thousands. Yields and coupons are in percent.
  - **Order quantities are not currency.** IBKR's quantity unit for bonds can be €1,000 of face value per unit (Bunds), so minimum sizes and order quantities must be converted before reading them as money.
  - **IBKR returns few bond details.** Ratings, issuer names, payment frequency and often currency, coupon and maturity come back empty. The scanner can filter on ratings but does not return them.
  - **Refinitiv data covers issuers, not bonds.** The balance-sheet ratio and ESG filters describe the issuing company, so every bond from the same issuer gets the same value.
- 

## 10. Looking up other codes

```
import xml.etree.ElementTree as ET

root = ET.parse("interactive_brokers_scanner_reference.xml").getroot()

# Filters available for an instrument
for instrument in root.iter("Instrument"):
    if instrument.findtext("type") == "BOND.GOV.T.NON-US":
        print(instrument.findtext("filters"))
        break

# Tag codes and allowed values for one filter id
for filter_list in root.findall("FilterList"):
    for f in filter_list:
        if f.findtext("id") == "BOND_PAYMENT_FREQ":
            for field in f.iter("AbstractField"):
                print(field.findtext("code"), [v.findtext("code") for v in
                    field.iter("ComboValue")])
```